

Flow Rate Incremental Controller 2.0

1 General Incremental Controller

Consider the controller output at sample k to be $u(k)$. In velocity-form PID control, the current output is updated from the previous output by adding an incremental change $\Delta u(k)$:

$$u(k) = u(k-1) + \Delta u(k)$$

PID equation

$$u(k) = K_p \cdot e(k) + K_i \cdot T_k \cdot \sum_{i=0}^k e(i) + K_d \cdot \frac{e(k) - e(k-1)}{T_k}$$

Where $e(k)$ is the error for a give sample and T_k is the controller loop time.

$$\Delta u(k) = u(k) - u(k-1)$$

$$\Delta u(k) = K_p \cdot (e(k) - e(k-1)) + K_i \cdot T_k \cdot \left(\sum_{i=0}^k e(i) - \sum_{i=0}^{k-1} e(i) \right) + K_d \cdot \left(\frac{e(k) - 2e(k-1) + e(k-2)}{T_k} \right)$$

$$\Delta u(k) = K_p \cdot (e(k) - e(k-1)) + K_i \cdot T_k \cdot e(k) + K_d \cdot \left(\frac{e(k) - 2e(k-1) + e(k-2)}{T_k} \right)$$

2 Flow Rate Control Application

PI incremental controller is adopted as the flow rate controller, as it provides a suitable balance between implementation simplicity, control responsiveness, and steady-state accuracy, while avoiding the increased noise sensitivity associated with derivative action.

2.1 Controller Constraints

Unconstrained PI incremental equation

$$u^*(k) = u(k-1) + \Delta u(k)$$

$$\Delta u(k) = \Delta u_P(k) + \Delta u_I(k)$$

$$u^*(k) = u(k-1) + \Delta u_P(k) + \Delta u_I(k)$$

$$\Delta u_P(k) = K_p \cdot (e(k) - e(k-1)) \quad , \quad \Delta u_I(k) = K_i \cdot T_k \cdot e(k)$$

$$u^*(k) = u(k-1) + K_p \cdot (e(k) - e(k-1)) + K_i \cdot T_k \cdot e(k)$$

To make this a *constrained* output, clamping must be added.

Constrained PI incremental equation

$$u(k) = \text{sat}[u^*(k)]$$

$$\text{sat}[u^*(k)] = \begin{cases} u_{\min} , & u^*(k) < u_{\min} \\ u^*(k) , & u_{\min} \leq u^*(k) \leq u_{\max} \\ u_{\max} , & u^*(k) > u_{\max} \end{cases}$$

For this specific case, $u_{\min} = 0$ and $u_{\max} = 100$.

2.2 Controller Anti-Windup

For anti-windup, if the controller is saturated, reject the delta term.

$$\Delta u(k) = \begin{cases} \Delta u_P(k) + \Delta u_I(k) , & \text{if } u^*(k) \text{ is within the output limits} \\ \Delta u_P(k) , & \text{if } u^*(k) \text{ exceeds the output limits} \end{cases}$$

$$\Delta u(k) = \begin{cases} \Delta u_P(k) + \Delta u_I(k) , & u(k) = u^*(k) \\ \Delta u_P(k) , & u(k) \neq u^*(k) \end{cases}$$